

```
# ----- #
# 0. SETUP
# ----- #

needed_packages <- c("ggplot2", "dplyr", "broom")

for (pkg in needed_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
}

library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.6.1

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.6.1

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(broom)
```

Warning: package 'broom' was built under R version 4.6.1

```
# Make printed numbers a little easier to read.
options(digits = 4)
```

```
# ===== #
# 1. Understanding the Data
# ===== #

# The original public source is:
# https://www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt
data_file <- "diabetes.tab.txt"
```

```

if (!file.exists(data_file)) {
  message("diabetes.tab.txt was not found in the current working directory.")
  message("Please choose the supplied diabetes.tab.txt file.")
  data_file <- file.choose()
}

# Read the supplied tab-separated file.
diabetes <- read.table(data_file, header = TRUE, sep = "\t")

# Convert column names to lower case so they are easier to type consistently.
names(diabetes) <- tolower(names(diabetes))

# The source dataset uses 'y' as the disease-progression outcome.
# Rename it to something clearer for the rest of the analysis.
if ("y" %in% names(diabetes)) {
  diabetes <- diabetes %>% rename(progression = y)
}

# first few rows.
head(diabetes)

```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	progression
1	59	2	32.1	101	157	93.2	38	4	4.860	87	151
2	48	1	21.6	87	183	103.2	70	3	3.892	69	75
3	72	2	30.5	93	156	93.6	41	4	4.673	85	141
4	24	1	25.3	84	198	131.4	40	5	4.890	89	206
5	50	1	23.0	101	192	125.4	52	4	4.290	80	135
6	23	1	22.6	89	139	64.8	61	2	4.190	68	97

```

# Number of rows and columns.
dim(diabetes)

```

```
[1] 442 11
```

```

# Variable names.
names(diabetes)

```

```

[1] "age"      "sex"      "bmi"      "bp"      "s1"
[6] "s2"      "s3"      "s4"      "s5"      "s6"
[11] "progression"

```

```

# Structure and data types.
str(diabetes)

```

```

'data.frame':  442 obs. of  11 variables:
 $ age      : num  59 48 72 24 50 23 36 66 60 29 ...
 $ sex      : num  2 1 2 1 1 1 2 2 2 1 ...
 $ bmi      : num  32.1 21.6 30.5 25.3 23 22.6 22 26.2 32.1 30 ...

```

```
$ bp      : num  101 87 93 84 101 89 90 114 83 85 ...
$ s1      : num  157 183 156 198 192 139 160 255 179 180 ...
$ s2      : num  93.2 103.2 93.6 131.4 125.4 ...
$ s3      : num  38 70 41 40 52 61 50 56 42 43 ...
$ s4      : num  4 3 4 5 4 2 3 4.55 4 4 ...
$ s5      : num  4.86 3.89 4.67 4.89 4.29 ...
$ s6      : num  87 69 85 89 80 68 82 92 94 88 ...
$ progression: num  151 75 141 206 135 97 138 63 110 310 ...
```

```
# The variable 'sex' is numerically coded in the source data, but it represents
# categories rather than a continuous measurement. Check the values present,
# then convert it to a factor so R treats it as categorical in later analyses.
unique(diabetes$sex)
```

```
[1] 2 1
```

```
diabetes$sex <- factor(diabetes$sex)

# Count missing values in each variable.
colSums(is.na(diabetes))
```

age	sex	bmi	bp	s1	s2
0	0	0	0	0	0
s3	s4	s5	s6	progression	
0	0	0	0	0	

```
# Total number of missing values.
sum(is.na(diabetes))
```

```
[1] 0
```

```
# Check for duplicated rows.
sum(duplicated(diabetes))
```

```
[1] 0
```

```
# Basic summary of every variable.
summary(diabetes)
```

age	sex	bmi	bp	s1
Min. :19.0	1:235	Min. :18.0	Min. : 62.0	Min. : 97
1st Qu.:38.2	2:207	1st Qu.:23.2	1st Qu.: 84.0	1st Qu.:164
Median :50.0		Median :25.7	Median : 93.0	Median :186
Mean :48.5		Mean :26.4	Mean : 94.6	Mean :189
3rd Qu.:59.0		3rd Qu.:29.3	3rd Qu.:105.0	3rd Qu.:210
Max. :79.0		Max. :42.2	Max. :133.0	Max. :301

s2	s3	s4	s5	s6
Min. : 41.6	Min. :22.0	Min. :2.00	Min. :3.26	Min. : 58.0

Statistic	1st Qu.	Median	Mean	3rd Qu.	Max.
1st Qu.:	96.0	113.0	115.4	134.5	242.4
Median :	40.2	48.0	49.8	57.8	99.0
Mean :	3.00	4.00	4.07	5.00	9.09
3rd Qu.:	4.28	4.62	4.64	5.00	6.11
Max. :	83.2	91.0	91.3	98.0	124.0

progression

Statistic	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Min. :	25	87	140	152	212	346

```
# FOR YOUR REPORT
# - What does one row represent?
# - Which variable is the outcome?
# - Which variables are predictors?
# - Which variables are numerical, and which are categorical in meaning?
# - Are there missing values, duplicated rows or obviously problematic values?
# - Why do health-related data require careful and responsible interpretation?
```

```
# ===== #
# QUESTION 2. EXPLORE THE DATA
# ===== #

# A helper function for common descriptive statistics.
describe_numeric <- function(x) {
  c(
    n = sum(!is.na(x)),
    mean = mean(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    sd = sd(x, na.rm = TRUE),
    min = min(x, na.rm = TRUE),
    q1 = quantile(x, 0.25, na.rm = TRUE),
    q3 = quantile(x, 0.75, na.rm = TRUE),
    max = max(x, na.rm = TRUE),
    iqr = IQR(x, na.rm = TRUE)
  )
}

# Descriptive statistics for the outcome.
describe_numeric(diabetes$progression)
```

	n	mean	median	sd	min	q1.25%	q3.75%	max	iqr
	442.00	152.13	140.50	77.09	25.00	87.00	211.50	346.00	124.50

```
# Descriptive statistics for BMI.
describe_numeric(diabetes$bmi)
```

	n	mean	median	sd	min	q1.25%	q3.75%	max	iqr
	442.000	26.376	25.700	4.418	18.000	23.200	29.275	42.200	6.075

```
# Descriptive statistics for age.
describe_numeric(diabetes$age)
```

	n	mean	median	sd	min	q1.25%	q3.75%	max	iqr
	442.00	48.52	50.00	13.11	19.00	38.25	59.00	79.00	20.75

```
# A compact descriptive table for all numeric variables.
# You do not need to put the whole table in your report.
```

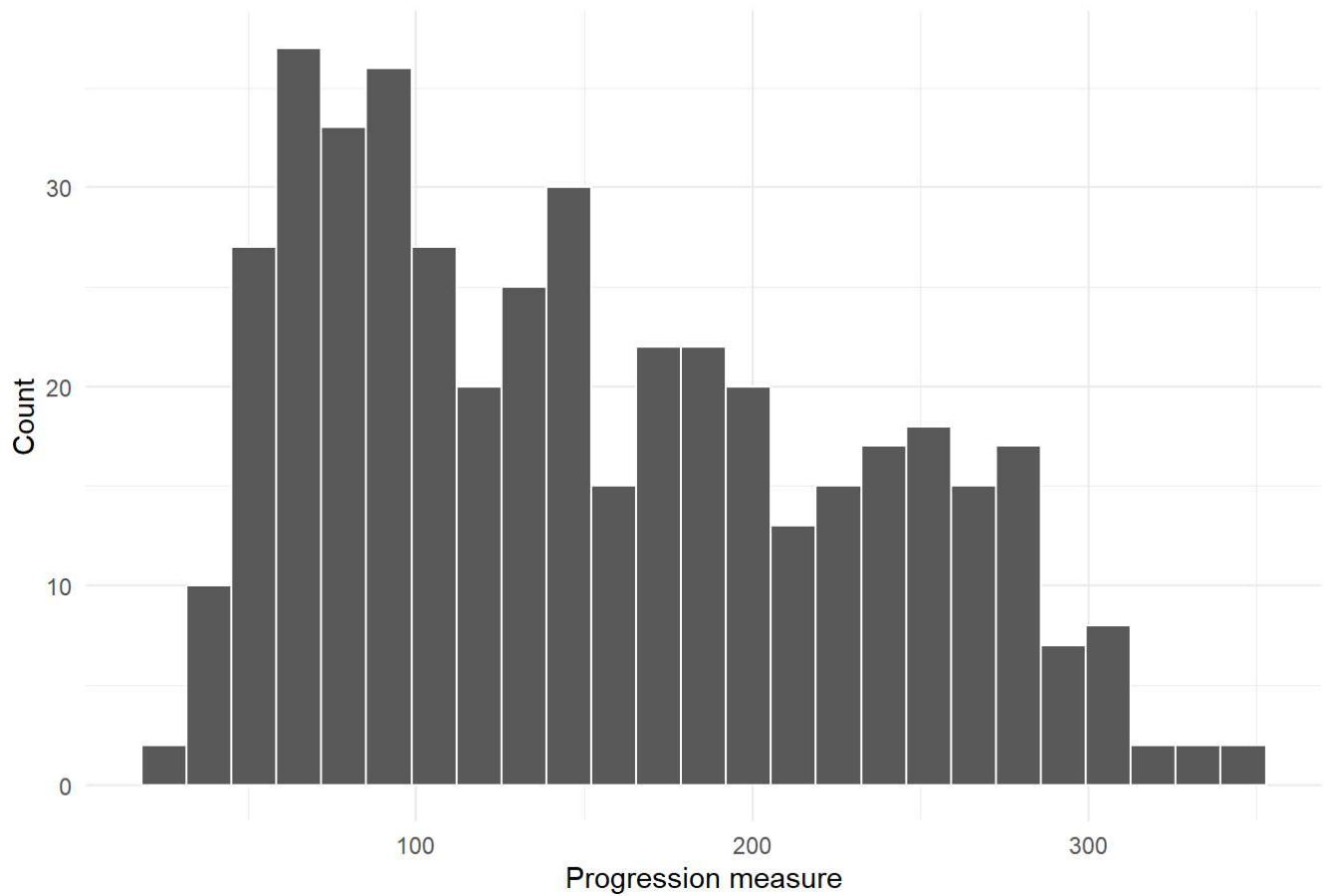
```
descriptive_table <- diabetes %>%
  summarise(across(
    where(is.numeric),
    list(
      mean = ~mean(.x, na.rm = TRUE),
      median = ~median(.x, na.rm = TRUE),
      sd = ~sd(.x, na.rm = TRUE),
      min = ~min(.x, na.rm = TRUE),
      max = ~max(.x, na.rm = TRUE)
    )
  ))

print(descriptive_table)
```

	age_mean	age_median	age_sd	age_min	age_max	bmi_mean	bmi_median	bmi_sd	bmi_min	
1	48.52	50	13.11	19	79	26.38	25.7	4.418	18	
	bmi_max	bp_mean	bp_median	bp_sd	bp_min	bp_max	s1_mean	s1_median	s1_sd	s1_min
1	42.2	94.65	93	13.83	62	133	189.1	186	34.61	97
	s1_max	s2_mean	s2_median	s2_sd	s2_min	s2_max	s3_mean	s3_median	s3_sd	s3_min
1	301	115.4	113	30.41	41.6	242.4	49.79	48	12.93	22
	s3_max	s4_mean	s4_median	s4_sd	s4_min	s4_max	s5_mean	s5_median	s5_sd	s5_min
1	99	4.07	4	1.29	2	9.09	4.641	4.62	0.5224	3.258
	s5_max	s6_mean	s6_median	s6_sd	s6_min	s6_max	progression_mean			
1	6.107	91.26	91	11.5	58	124	152.1			
	progression_median	progression_sd	progression_min	progression_max						
1		140.5	77.09	25			346			

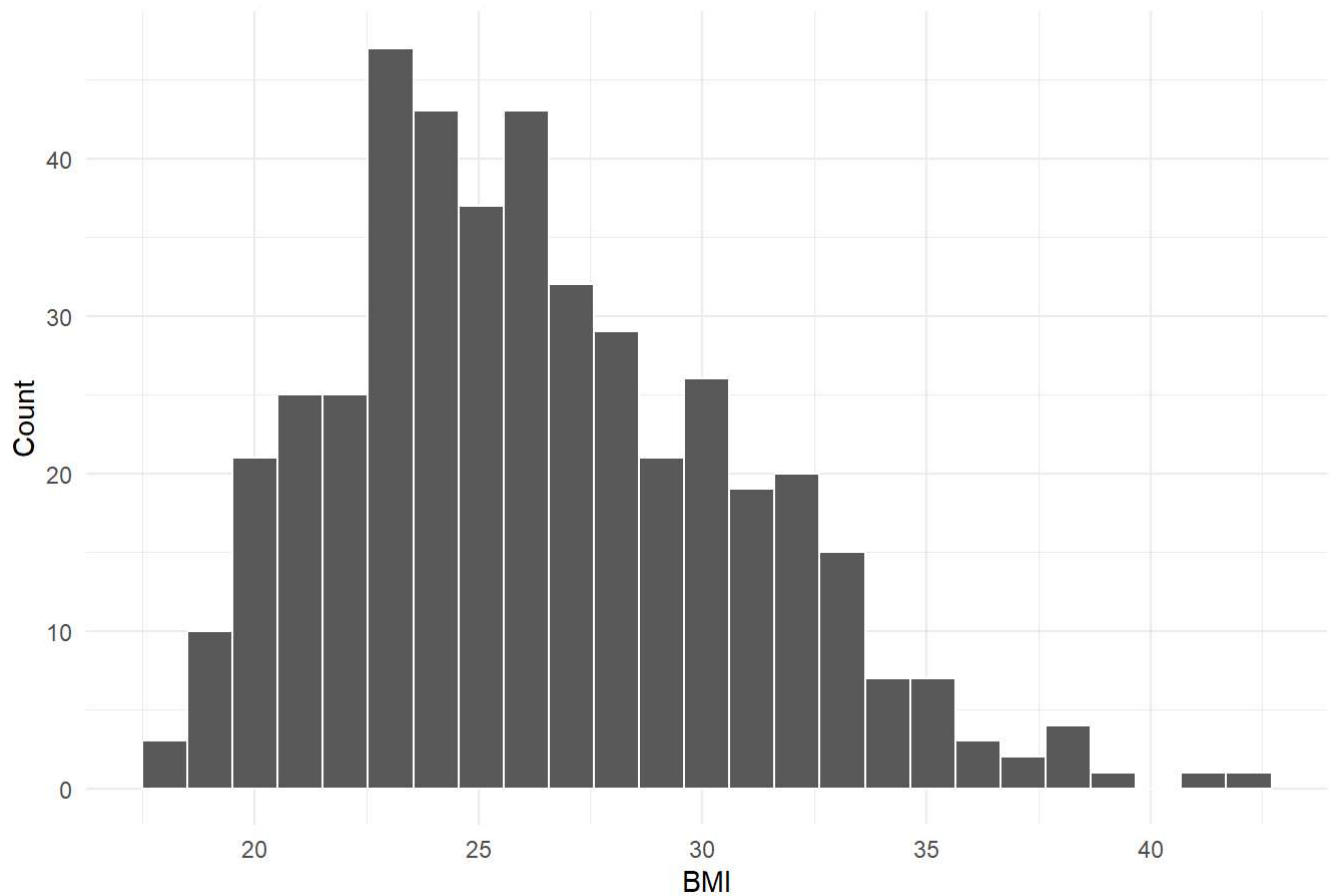
```
# Distribution of diabetes progression.
ggplot(diabetes, aes(x = progression)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of diabetes progression",
    x = "Progression measure",
    y = "Count"
  ) +
  theme_minimal()
```

Distribution of diabetes progression



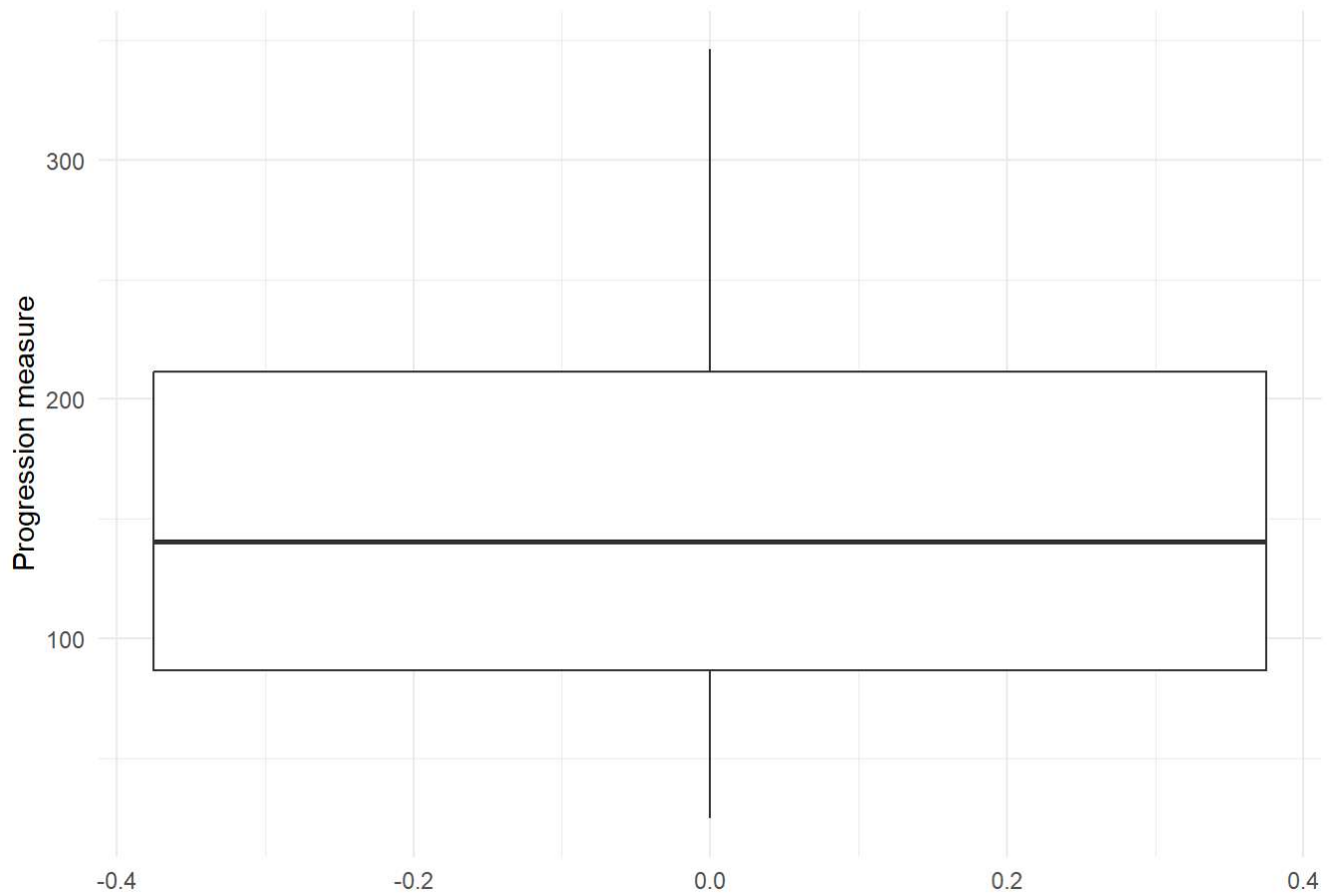
```
# Distribution of BMI.
ggplot(diabetes, aes(x = bmi)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of BMI",
    x = "BMI",
    y = "Count"
  ) +
  theme_minimal()
```

Distribution of BMI

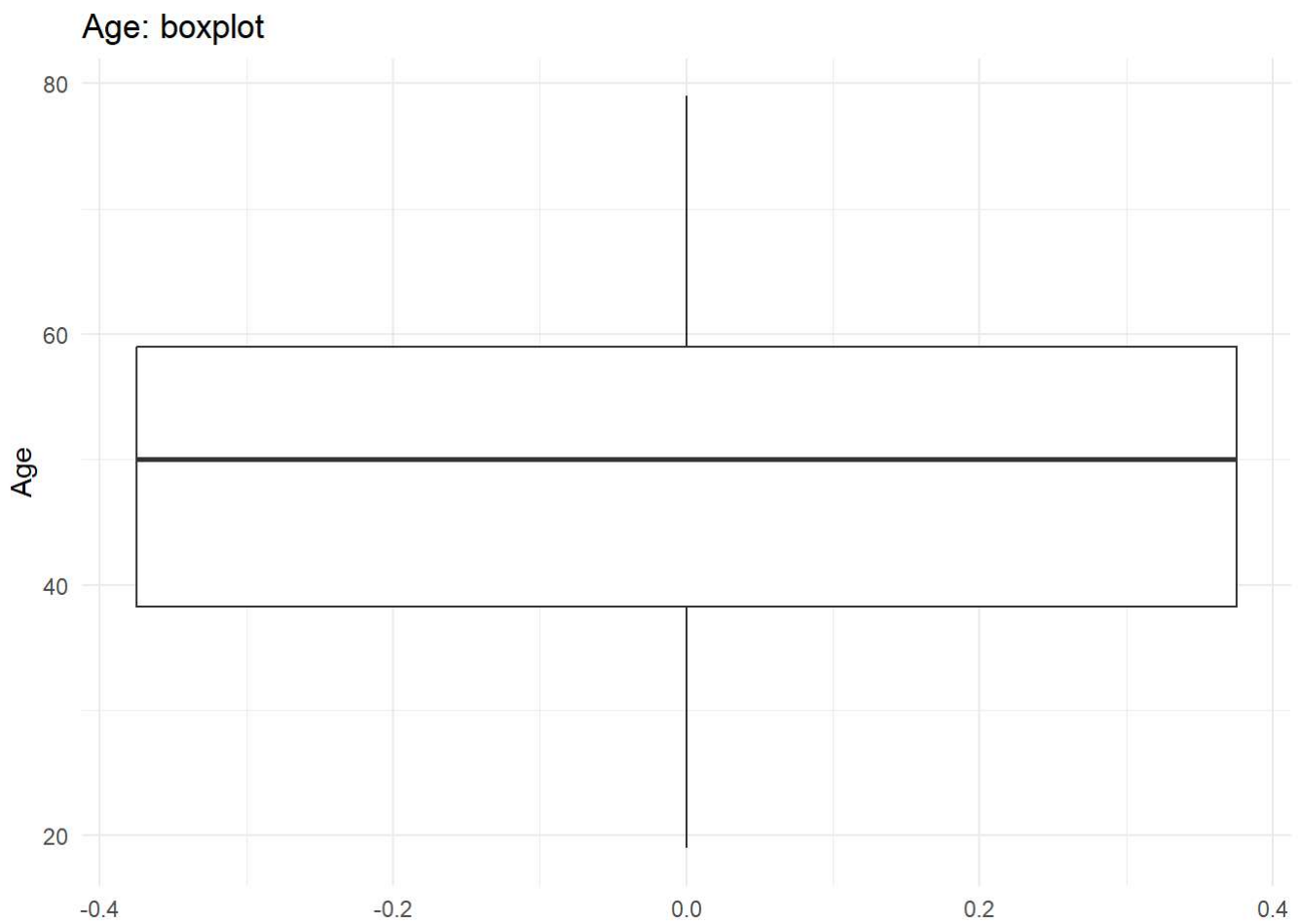


```
# One additional useful visualisation: boxplot of progression.
ggplot(diabetes, aes(y = progression)) +
  geom_boxplot() +
  labs(
    title = "Diabetes progression: boxplot",
    y = "Progression measure",
    x = NULL
  ) +
  theme_minimal()
```

Diabetes progression: boxplot

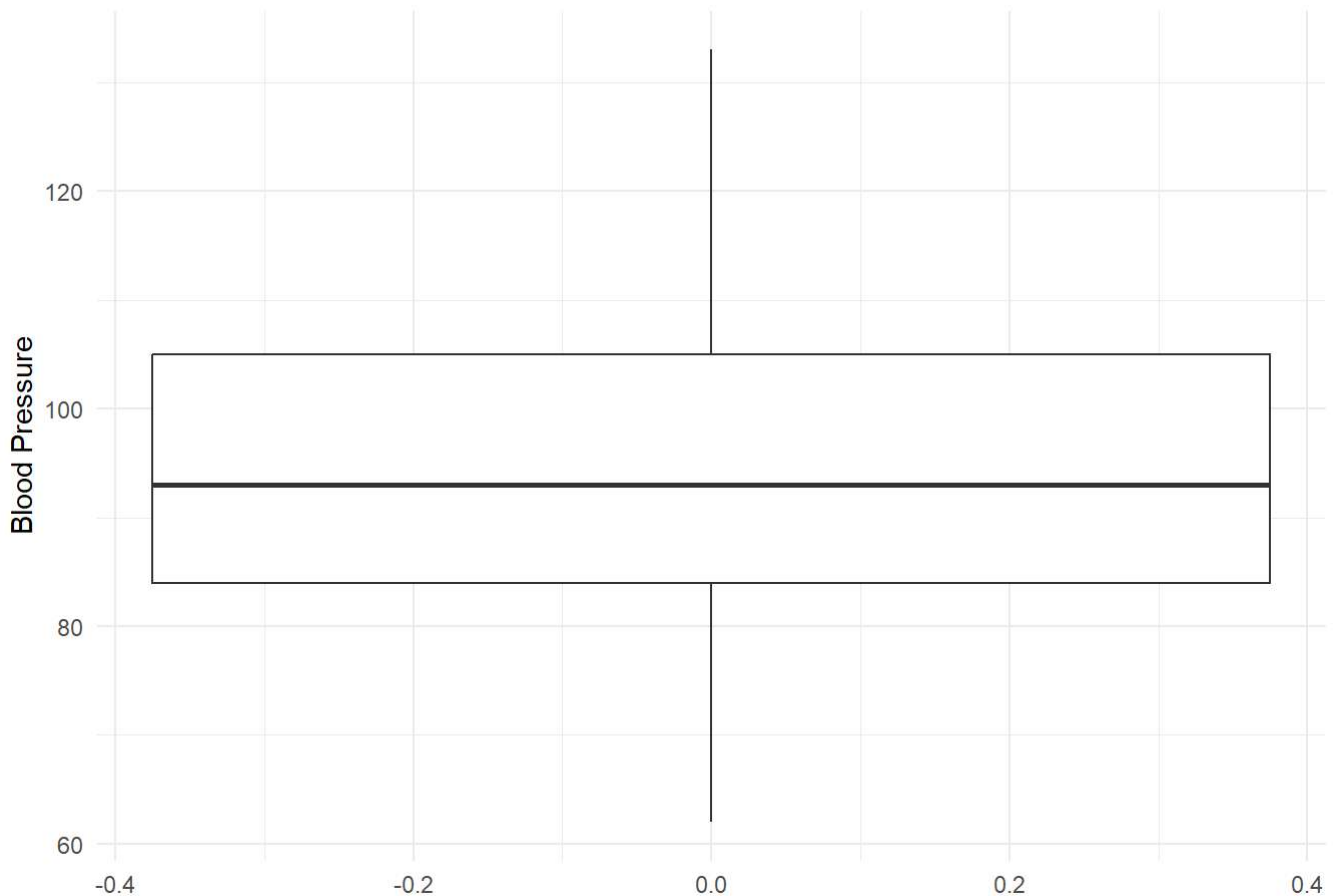


```
ggplot(diabetes, aes(y = age)) +  
  geom_boxplot() +  
  labs(  
    title = "Age: boxplot",  
    y = "Age",  
    x = NULL  
  ) +  
  theme_minimal()
```



```
ggplot(diabetes, aes(y = bp)) +  
  geom_boxplot() +  
  labs(  
    title = "Blood Pressure: boxplot",  
    y = "Blood Pressure",  
    x = NULL  
  ) +  
  theme_minimal()
```

Blood Pressure: boxplot



```
# FOR YOUR REPORT
# - What do the centre and spread of the main variables look like?
# - Are the distributions approximately symmetric or clearly skewed?
# - Are there unusual observations?
# - Which features of the data matter before modelling begins?
```

```
# ===== #
# QUESTION 3. INVESTIGATE RELATIONSHIPS
# ===== #

# Keep only genuinely numerical measurement variables for Pearson correlations.
# 'sex' has already been converted to a factor, so it is excluded automatically.
numeric_data <- diabetes %>% select(where(is.numeric))

# Correlation matrix.
cor_matrix <- cor(numeric_data, use = "complete.obs", method = "pearson")
round(cor_matrix, 2)
```

	age	bmi	bp	s1	s2	s3	s4	s5	s6	progression
age	1.00	0.19	0.34	0.26	0.22	-0.08	0.20	0.27	0.30	0.19
bmi	0.19	1.00	0.40	0.25	0.26	-0.37	0.41	0.45	0.39	0.59
bp	0.34	0.40	1.00	0.24	0.19	-0.18	0.26	0.39	0.39	0.44

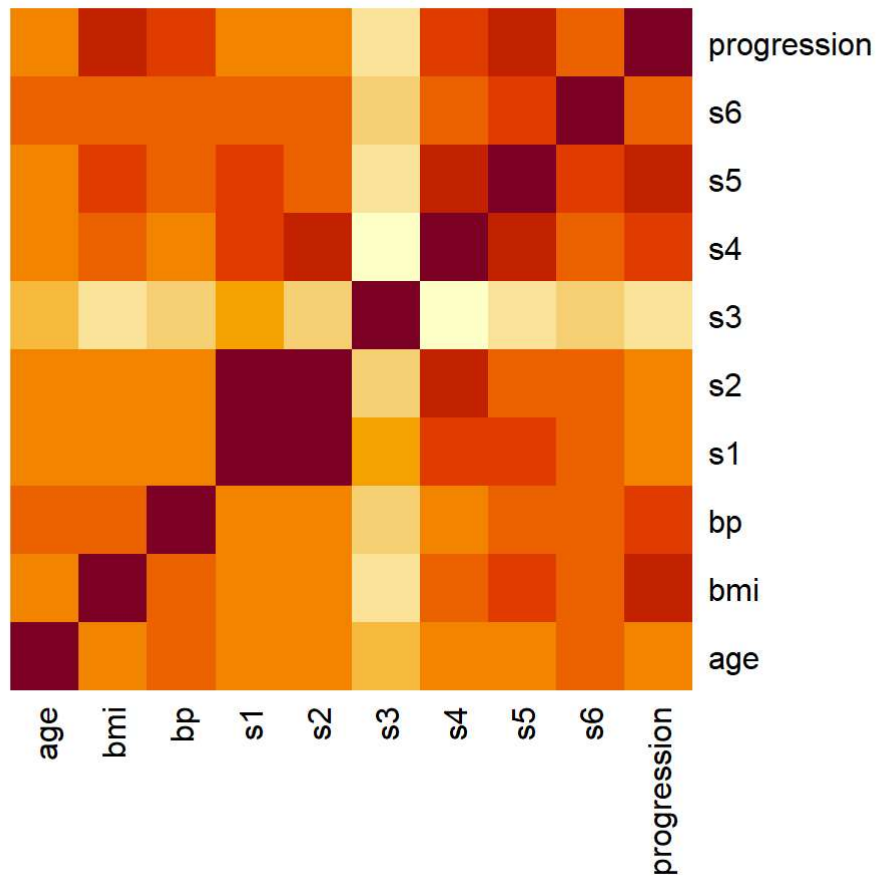
s1	0.26	0.25	0.24	1.00	0.90	0.05	0.54	0.52	0.33	0.21
s2	0.22	0.26	0.19	0.90	1.00	-0.20	0.66	0.32	0.29	0.17
s3	-0.08	-0.37	-0.18	0.05	-0.20	1.00	-0.74	-0.40	-0.27	-0.39
s4	0.20	0.41	0.26	0.54	0.66	-0.74	1.00	0.62	0.42	0.43
s5	0.27	0.45	0.39	0.52	0.32	-0.40	0.62	1.00	0.46	0.57
s6	0.30	0.39	0.39	0.33	0.29	-0.27	0.42	0.46	1.00	0.38
progression	0.19	0.59	0.44	0.21	0.17	-0.39	0.43	0.57	0.38	1.00

```
# Correlations of all numeric variables with progression.
# The outcome itself will appear with correlation 1, so focus on the predictors.
progression_correlations <- sort(
  cor_matrix[, "progression"],
  decreasing = TRUE
)
print(progression_correlations)
```

progression	bmi	s5	bp	s4	s6
1.0000	0.5865	0.5659	0.4415	0.4305	0.3825
s1	age	s2	s3		
0.2120	0.1879	0.1741	-0.3948		

```
# Simple heatmap using base R.
# Rowv = NA and Colv = NA keep the variables in their original order instead
# of reordering them with clustering dendrograms.
heatmap(
  cor_matrix,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  symm = TRUE,
  margins = c(7, 7),
  main = "Correlation heatmap"
)
```

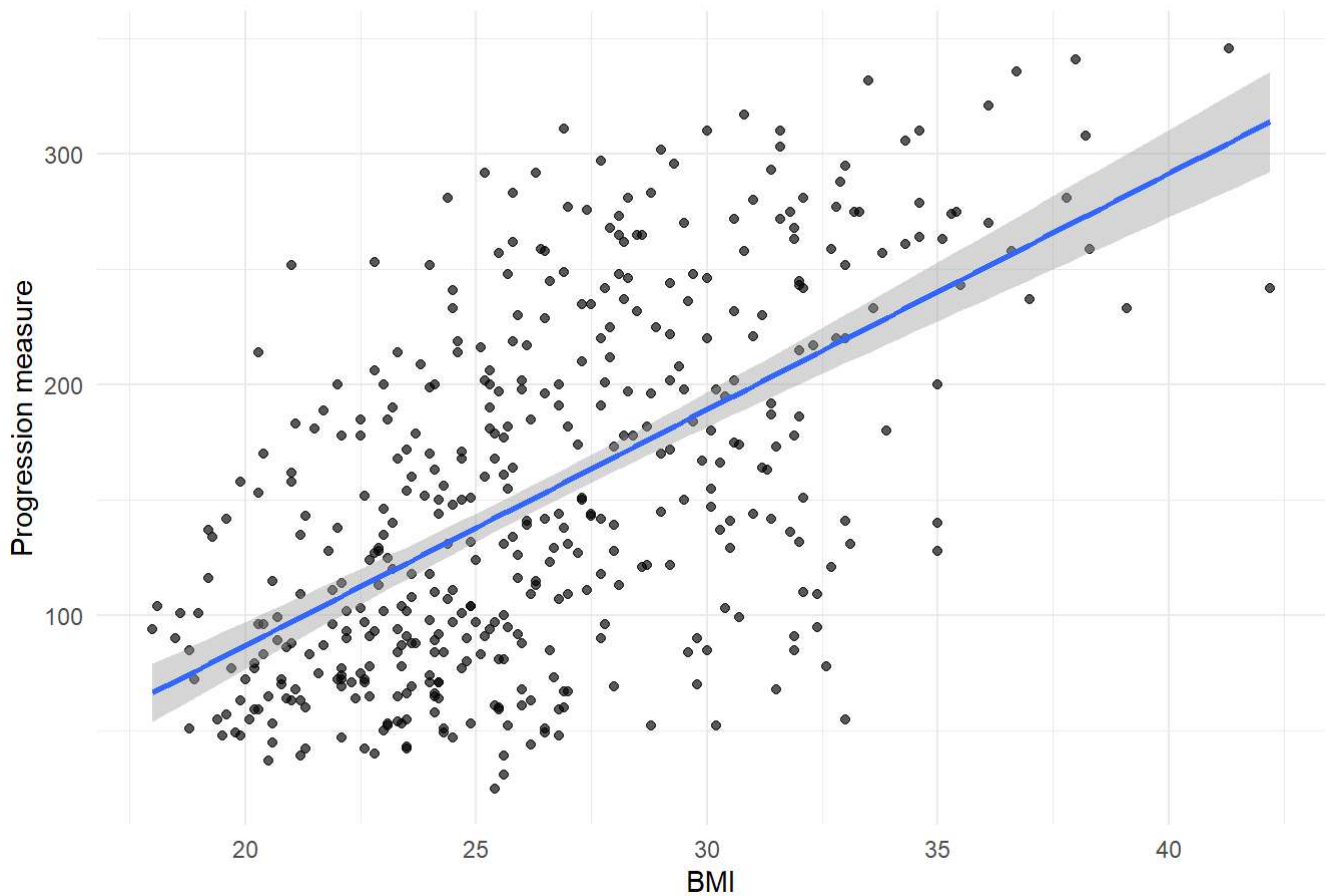
Correlation heatmap



```
# Scatterplot 1: BMI and progression.
ggplot(diabetes, aes(x = bmi, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus BMI",
    x = "BMI",
    y = "Progression measure"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

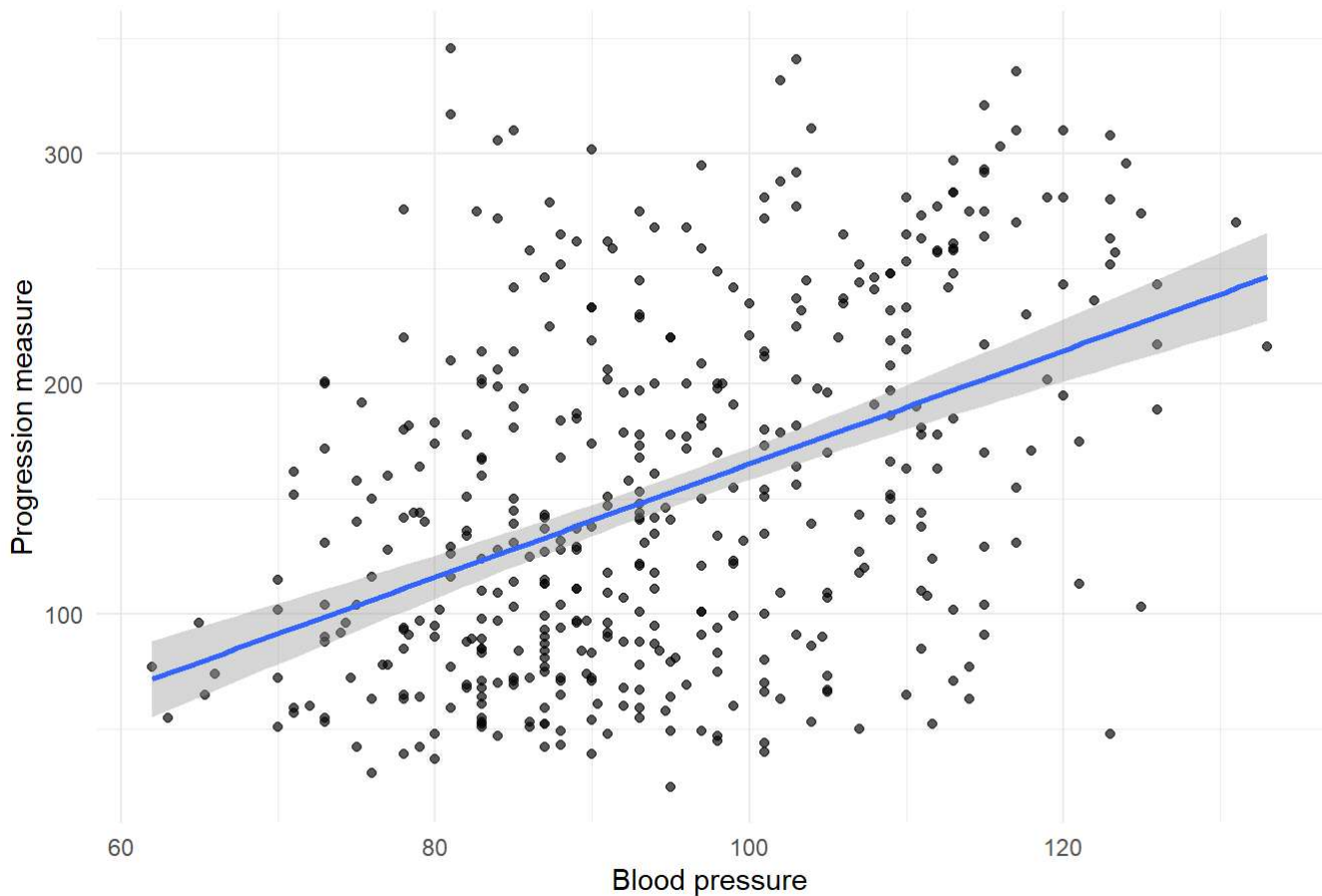
Progression versus BMI



```
# Scatterplot 2: blood pressure and progression.
ggplot(diabetes, aes(x = bp, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus blood pressure",
    x = "Blood pressure",
    y = "Progression measure"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Progression versus blood pressure



```
# FOR YOUR REPORT
# - Which predictors are most strongly associated with progression?
# - Which predictors are strongly related to each other?
# - Could some predictors carry overlapping information?
# - Which variables look most promising for regression analysis, and why?
```

```
# ===== #
# QUESTION 4. BUILD AND INTERPRET REGRESSION MODELS
# ===== #

# ---- Simple linear regression -----

# BMI is used as the starting predictor because it is meaningful and commonly
# examined in this dataset. You may choose another predictor if you justify it.
simple_model <- lm(progression ~ bmi, data = diabetes)

# Standard model summary.
summary(simple_model)
```

Call:

```
lm(formula = progression ~ bmi, data = diabetes)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-164.92	-43.57	-8.65	46.34	154.88

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-117.773	18.019	-6.54	1.8e-10 ***
bmi	10.233	0.674	15.19	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 62.5 on 440 degrees of freedom

Multiple R-squared: 0.344, Adjusted R-squared: 0.342

F-statistic: 231 on 1 and 440 DF, p-value: <2e-16

Tidy coefficient table, including confidence intervals.

```
tidy(simple_model, conf.int = TRUE)
```

A tibble: 2 × 7

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-118.	18.0	-6.54	1.75e-10	-153.	-82.4
2	bmi	10.2	0.674	15.2	3.47e-42	8.91	11.6

Model-level statistics such as R-squared and adjusted R-squared.

```
glance(simple_model)
```

A tibble: 1 × 12

	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.344	0.342	62.5	231.	3.47e-42	1	-2454.	4914.	4926.

i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

Coefficients for writing the fitted equation.

```
coef(simple_model)
```

(Intercept)	bmi
-117.77	10.23

FOR YOUR REPORT

- Write the fitted equation.

- Interpret the slope in context.

- Does the intercept have a useful real-world interpretation?

- What does R-squared mean here?

---- Multiple regression -----

```
# This is a ready-to-run starting model using three predictors.
# You may change the predictor set after examining your EDA and correlations.
multiple_model <- lm(
  progression ~ bmi + bp + s5,
  data = diabetes
)

summary(multiple_model)
```

Call:

```
lm(formula = progression ~ bmi + bp + s5, data = diabetes)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-140.23	-40.64	-2.19	38.27	139.80

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-334.881	25.878	-12.94	< 2e-16 ***
bmi	6.500	0.697	9.32	< 2e-16 ***
bp	0.903	0.217	4.17	3.7e-05 ***
s5	49.577	5.890	8.42	5.6e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55.8 on 438 degrees of freedom

Multiple R-squared: 0.48, Adjusted R-squared: 0.477

F-statistic: 135 on 3 and 438 DF, p-value: <2e-16

```
tidy(multiple_model, conf.int = TRUE)
```

A tibble: 4 × 7

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-335.	25.9	-12.9	1.16e-32	-386.	-284.
2	bmi	6.50	0.697	9.32	5.55e-19	5.13	7.87
3	bp	0.903	0.217	4.17	3.74e- 5	0.477	1.33
4	s5	49.6	5.89	8.42	5.56e-16	38.0	61.2

```
glance(multiple_model)
```

A tibble: 1 × 12

	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.480	0.477	55.8	135.	7.16e-62	3	-2403.	4815.	4836.

i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>

```

# Compare the simple and multiple models using measures taught in the course.
# R-squared shows explained variation, while adjusted R-squared takes model
# complexity into account when several predictors are used.
model_comparison <- bind_rows(
  glance(simple_model) %>% mutate(model = "Simple: BMI"),
  glance(multiple_model) %>% mutate(model = "Multiple: BMI + BP + S5")
) %>%
  select(model, r.squared, adj.r.squared, sigma)

print(model_comparison)

```

```

# A tibble: 2 × 4
  model                r.squared adj.r.squared sigma
  <chr>                <dbl>      <dbl> <dbl>
1 Simple: BMI          0.344        0.342  62.5
2 Multiple: BMI + BP + S5 0.480        0.477  55.8

```

```

# FOR YOUR REPORT
# - Interpret at least two coefficients from the multiple model.
# - Use the phrase 'holding the other predictors constant' correctly.
# - Compare the simple and multiple models.
# - Do not assume that the model with the larger R-squared is automatically
#   better in every respect.

```

```

# ===== #
# QUESTION 5. EVALUATE THE ANALYSIS
# ===== #

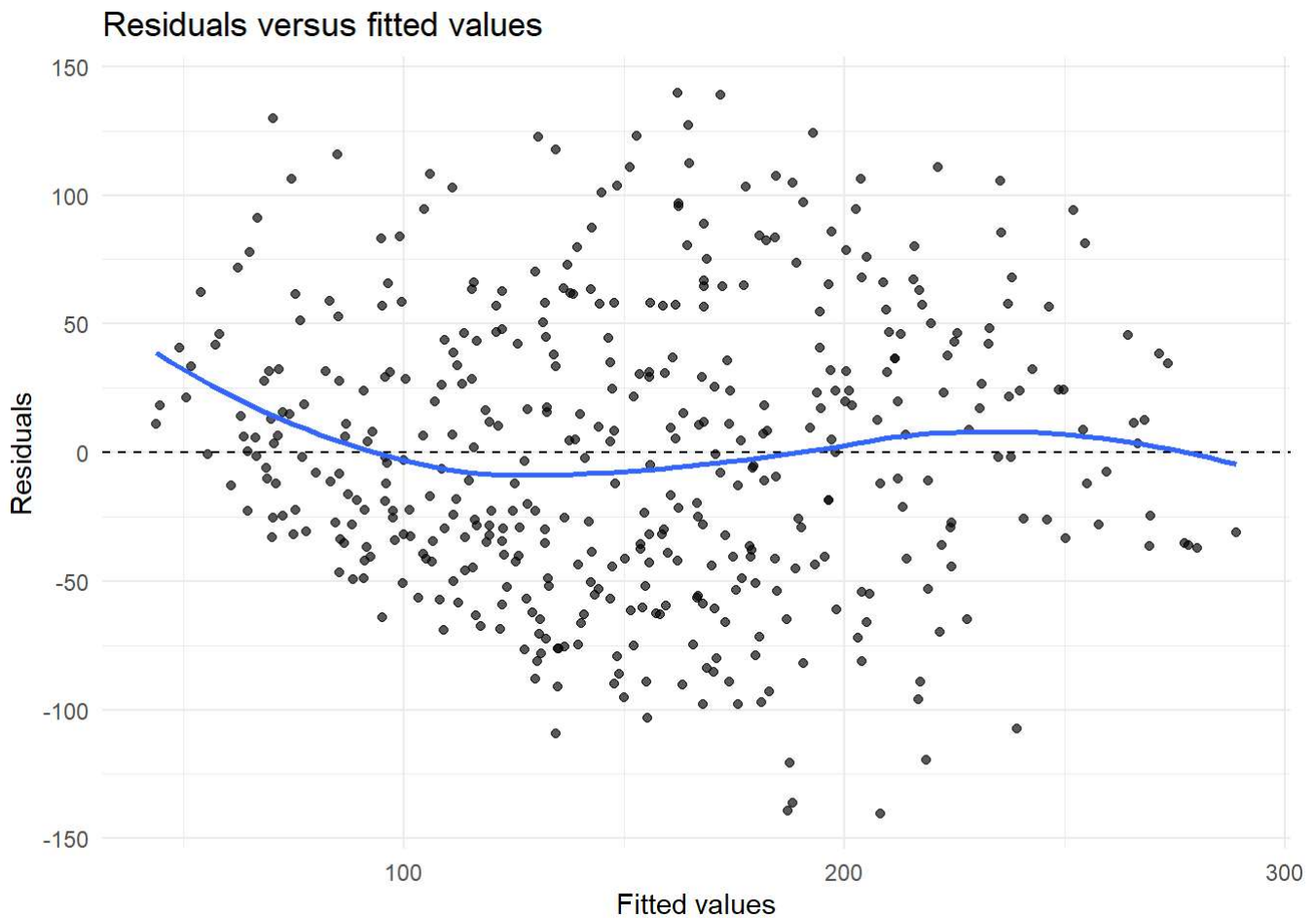
# ---- Residuals versus fitted values -----

# augment() creates fitted values and residuals in a convenient data frame.
diagnostic_data <- augment(multiple_model)

ggplot(diagnostic_data, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.65) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_smooth(se = FALSE) +
  labs(
    title = "Residuals versus fitted values",
    x = "Fitted values",
    y = "Residuals"
  ) +
  theme_minimal()

```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



```
# Look for obvious structure rather than trying to prove that every assumption
# is perfectly satisfied. A strong curve or a clear change in spread would be
# warning signs for a simple linear model.
```

```
# ---- Multicollinearity using VIF -----
```

```
# VIF can be calculated without installing another package.
```

```
manual_vif <- function(model) {
  X <- model.matrix(model)[, -1, drop = FALSE] # remove intercept

  vif_values <- sapply(seq_len(ncol(X)), function(i) {
    target <- X[, i]
    others <- X[, -i, drop = FALSE]

    if (ncol(others) == 0) {
      return(1)
    }

    r2 <- summary(lm(target ~ others))$r.squared
    1 / (1 - r2)
  })

  setNames(vif_values, colnames(X))
}
```

```
}
```

```
manual_vif(multiple_model)
```

```
      bmi      bp      s5  
1.345 1.274 1.342
```

```
# There is no single universal VIF cut-off. Interpret the values together with  
# the correlation matrix and with the purpose of the model.
```

```
# ---- Statistical uncertainty -----
```

```
# A tidy table with estimates, standard errors, t-values, p-values and 95%  
# confidence intervals.
```

```
coef_table <- tidy(multiple_model, conf.int = TRUE)  
print(coef_table)
```

```
# A tibble: 4 × 7
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	-335.	25.9	-12.9	1.16e-32	-386.	-284.
2 bmi	6.50	0.697	9.32	5.55e-19	5.13	7.87
3 bp	0.903	0.217	4.17	3.74e- 5	0.477	1.33
4 s5	49.6	5.89	8.42	5.56e-16	38.0	61.2

```
# You can also obtain the confidence intervals directly.
```

```
confint(multiple_model, level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	-385.7426	-284.020
bmi	5.1300	7.870
bp	0.4769	1.329
s5	38.0002	61.154

```
# FOR YOUR REPORT
```

```
# - Comment on the residuals-versus-fitted plot.
```

```
# - Use VIF and the correlation matrix to discuss multicollinearity.
```

```
# - For at least two important coefficients, discuss the estimate and either  
# the confidence interval, t-value or p-value.
```

```
# - Distinguish statistical significance from practical importance.
```

```
# - State the main statistical weaknesses or uncertainties in the analysis.
```